Unit-I

Definition

- *Machine Design is the creation of new and bette existing ones.*
- *A new or better machine is one which is more econ production and operation.*
- *The process of design is a long and time consuming. From existing, a new idea has to be conceived. The idea is then studied in mind its commercial success and given shap drawings.*
- *In the preparation of these drawings, care must be taken of the av resources in money, in men and in materials required f completion of the new idea into an actual realit*
- *In designing a machine, it is necessary to have a good many subjects such as Mathematics, Engineering Mechanics, Strength of Materials, Theory of Machines, Workshop Processes and Engineer*

Classifications of Machine Design

The machine design may be classified as follows:

1. Adaptive design. *In most cases, the designer's work is concerned adaptation of existing designs. This type of design needs no specia and can be attempted by designers. The designer only makes minor alternation or modification in the exi*

2. Development design. *This type of design needs considera and design ability in order to modify the existing new material or different method. In this case, though the designer starts from the existing design, but the final product may differ qu original product.*

3. New design. *This type of design needs lot of research, technical ability and creative thinking.*

Only those designers who have personal qualities o up the work of a new design.

The designs, depending upon the methods used, may be classified as follo *(a)* ***Rational design.*** *This type of design depends upon mathematical formulae of principle of mechanics.*

(b) ***Empirical design.*** *This type of design depends upon empirical formulae based on the practice and past experience.*

(c) ***Industrial design.*** *This type of design depends upon the production aspects to manufacture any machine component in the industry.*

(d) ***Optimum design.*** *It is the best design for the given objective function under the specified constraints. It may be achieved by minimizing the undesirable effects.*

(e) ***System design.*** *It is the design of any complex mechanical system like a motor car.*

(f) ***Element design.*** *It is the design of any element of the mechanical system like piston, crankshaft, connecting rod, etc.*

(g) ***Computer aided design.*** *This type of design depends upon the use of computer systems to assist in the creation, modification, analysis and optimization of a design.*

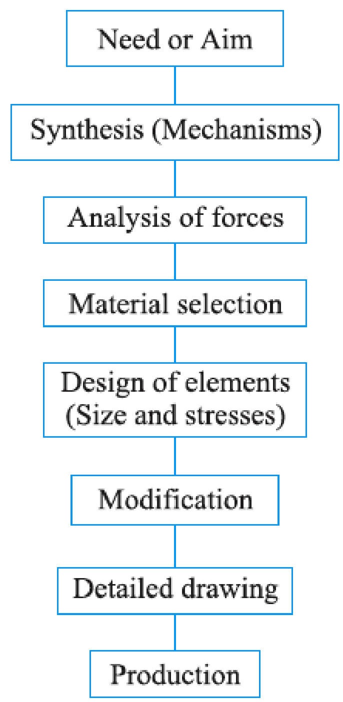
General Considerations in Machine Design

- ✓ *Type of load and stresses caused by the load*
- ✓ *Motion of the parts or kinematics of the machine*
- ✓ *Selection of materials*
- ✓ *Form and size of the parts.*
- ✓ *Frictional resistance and lubrication*
- ✓ *Convenient and economical features*
- ✓ *Use of standard parts*
- ✓ *Safety of operation*
- ✓ *Workshop facilities*
- ✓ *Cost of construction*
- ✓ *Assembling*

General Procedure in Machine Design

In designing a machine component, there is no rigid rule. The problem may be attempted in several ways. However, the general procedure to solve a design problem is as follows:

- ✓ ***Recognition of need.*** *First of all, make a complete statement of the problem, indicating the need, aim or purpose for which the machine is to be designed.*
- ✓ ***Synthesis (Mechanisms).*** *Select the possible mechanism or group of mechanisms which will give the desired motion.*
- ✓ ***Analysis of forces.*** *Find the forces acting on each member of the machine and the energy transmitted by each member.*
- ✓ ***Material selection.*** *Select the material best suited for each member of the machine.*



General procedure in Machine Design.

- ✓ ***Design of elements (Size and Stresses).*** *Find the size of each member of the machine by considering the force acting on the member and the permissible stresses for the material used. It should be kept in mind that each member should not deflect or deform than the permissible limit.*
- ✓ ***Modification.*** *Modify the size of the member to agree with the past experience and judgment to facilitate manufacture. The modification may*

also be necessary by consideration of manufacturing to reduce overall cost.

- ✓ ***Detailed drawing.*** *Draw the detailed drawing of each component and the assembly of the machine with complete specification for the manufacturing processes suggested.*
- ✓ ***Production.*** *The component, as per the drawing, is manufactured in the workshop.*

Engineering Materials

- ✓ *Choice of materials for a machine element depends very much on its properties, cost, availability and such other factors.*
- ✓ *It is therefore important to have some idea of the common engineering materials and their properties before learning the details of design procedure.*

Classification of Engineering Materials

The engineering materials are mainly classified as:

1. *Metals and their alloys, such as iron, steel, copper, aluminium, etc.*

2. *Non-metals, such as glass, rubber, plastic, etc.*

The metals may be further classified as:

(a) Ferrous metals and (b) Non-ferrous metals.

*The *ferrous metals are those which have the iron as their main constituent, such as cast iron, wrought iron and steel.*

The non-ferrous metals are those which have a metal other than iron as their main constituent, such as copper, aluminium, brass, tin, zinc, etc.

Selection of Materials for Engineering Purposes

The selection of a proper material, for engineering purposes, is one of the most difficult problems for the designer. The best material is one which serves the desired objective at the minimum cost. The following factors should be considered while selecting the material:

- ✓ *Availability of the materials,*
- ✓ *Suitability of the materials for the working conditions in service, and*

- ✓ *The cost of the materials.*

The stress in a plane which is at right angles to the line of action of the force. But it has been observed that at any point in a strained material, there are three planes, mutually perpendicular to each other which carry direct stresses only and no shear stress. It may be noted that out of these three direct stresses, one will be maximum and the other will be minimum. These perpendicular planes which have no shear stress are known as ***principal planes*** *and the direct stresses along these planes are known as* ***principal stresses.*** *The planes on which the maximum shear stress act are known as planes of maximum shear.*

A hollow shaft of 40 mm outer diameter and 25 mm inner diameter is subjected to a twisting moment of 120 N-m, simultaneously; it is subjected to an axial thrust of 10 kN and a bending moment of 80 N-m. Calculate the maximum compressive and shear stresses.

Solution. *Given: do = 40 mm; di = 25 mm; T = 120 N-m = 120 × 103 N-mm; P = 10 kN*

= 10 × 103 N; M = 80 N-m = 80 × 103 N-mm

We know that cross-sectional area of the shaft,

$$A = \frac{\pi}{4}\left[(d_o)^2 - (d_i)^2\right] = \frac{\pi}{4}\left[(40)^2 - (25)^2\right] = 766 \text{ mm}^2$$

∴ Direct compressive stress due to axial thrust,

$$\sigma_o = \frac{P}{A} = \frac{10 \times 10^3}{766} = 13.05 \text{ N/mm}^2 = 13.05 \text{ MPa}$$

Section modulus of the shaft,

$$Z = \frac{\pi}{32}\left[\frac{(d_o)^4 - (d_i)^4}{d_o}\right] = \frac{\pi}{32}\left[\frac{(40)^4 - (25)^4}{40}\right] = 5325 \text{ mm}^3$$

∴ Bending stress due to bending moment,

$$\sigma_b = \frac{M}{Z} = \frac{80 \times 10^3}{5325} = 15.02 \text{ N/mm}^2 = 15.02 \text{ MPa (compressive)}$$

and resultant compressive stress,

$$\sigma_c = \sigma_b + \sigma_o = 15.02 + 13.05 = 28.07 \text{ N/mm}^2 = 28.07 \text{ MPa}$$

We know that twisting moment (T),

$$120 \times 10^3 = \frac{\pi}{16} \times \tau\left[\frac{(d_o)^4 - (d_i)^4}{d_o}\right] = \frac{\pi}{16} \times \tau\left[\frac{(40)^4 - (25)^4}{40}\right] = 10\,650\ \tau$$

$$\therefore \quad \tau = 120 \times 10^3/10\,650 = 11.27 \text{ N/mm}^2 = 11.27 \text{ MPa}$$

Maximum compressive stress

We know that maximum compressive stress,

$$\sigma_{c(max)} = \frac{\sigma_c}{2} + \frac{1}{2}\left[\sqrt{(\sigma_c)^2 + 4\,\tau^2}\right]$$

$$= \frac{28.07}{2} + \frac{1}{2}\left[\sqrt{(28.07)^2 + 4\,(11.27)^2}\right]$$

$$= 14.035 + 18 = 32.035 \text{ MPa } \textbf{Ans.}$$

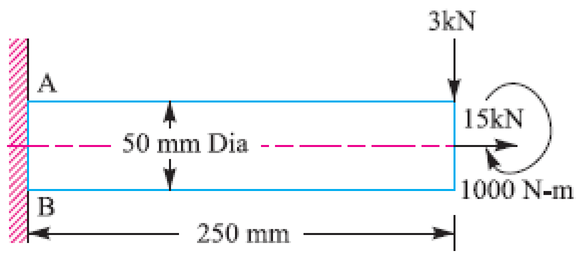
Maximum shear stress

We know that maximum shear stress,

$$\tau_{max} = \tfrac{1}{2}\left[\sqrt{(\sigma_c)^2 + 4\tau^2}\right] = \tfrac{1}{2}\left[\sqrt{(28.07)^2 + 4(11.27)^2}\right] = 18 \text{ MPa}$$

A shaft, as shown in Fig. 5.17, is subjected to a bending load of 3 kN, pure torque of 1000 N-m and an axial pulling force of 15 kN. Calculate the stresses at A and B.

Given: W = 3 kN = 3000 N; T = 1000 N-m = 1 × 106 N-mm; P = 15 kN = 15 × 103 N; d = 50 mm; x = 250 mm



We know that cross-sectional area of the shaft

$$A = \frac{\pi}{4} \times d^2$$

$$= \frac{\pi}{4}\,(50)^2 = 1964 \text{ mm}^2$$

∴ *Tensile stress due to axial pulling at points A and B,*

$$\sigma_o = \frac{P}{A} = \frac{15 \times 10^3}{1964} = 7.64 \text{ N/mm}^2 = 7.64 \text{ MPa}$$

Bending moment at points A and B,

$$M = W.x = 3000 \times 250 = 750 \times 10^3 \text{ N-mm}$$

Section modulus for the shaft,

$$Z = \frac{\pi}{32} \times d^3 = \frac{\pi}{32}\ (50)^3$$

$$= 12.27 \times 10^3 \text{ mm}^3$$

∴ Bending stress at points A and B,

$$\sigma_b = \frac{M}{Z} = \frac{750 \times 10^3}{12.27 \times 10^3}$$

$$= 61.1 \text{ N/mm}^2 = 61.1 \text{ MPa}$$

This bending stress is tensile at point A and compressive at point B.

∴ Resultant tensile stress at point A,

$$\sigma_A = \sigma_b + \sigma_o = 61.1 + 7.64$$
$$= 68.74 \text{ MPa}$$

and resultant compressive stress at point B,

$$\sigma_B = \sigma_b - \sigma_o = 61.1 - 7.64 = 53.46 \text{ MPa}$$

We know that the shear stress at points A and B due to the torque transmitted,

$$\tau = \frac{16\ T}{\pi\ d^3} = \frac{16 \times 1 \times 10^6}{\pi\ (50)^3} = 40.74 \text{ N/mm}^2 = 40.74 \text{ MPa} \qquad \ldots\left(\because\ T = \frac{\pi}{16} \times \tau \times d^3\right)$$

Stresses at point A

We know that maximum principal (or normal) stress at point A,

$$\sigma_{A(max)} = \frac{\sigma_A}{2} + \frac{1}{2}\left[\sqrt{(\sigma_A)^2 + 4\ \tau^2}\right]$$
$$= \frac{68.74}{2} + \frac{1}{2}\left[\sqrt{(68.74)^2 + 4\ (40.74)^2}\right]$$
$$= 34.37 + 53.3 = 87.67 \text{ MPa (tensile)}$$

Minimum principal (or normal) stress at point A,

$$\sigma_{A(min)} = \frac{\sigma_A}{2} - \frac{1}{2}\left[\sqrt{(\sigma_A)^2 + 4\ \tau^2}\right] = 34.37 - 53.3 = -18.93 \text{ MPa}$$
$$= 18.93 \text{ MPa (compressive)}$$

and maximum shear stress at point A,

$$\tau_{A(max)} = \tfrac{1}{2}\left[\sqrt{(\sigma_A)^2 + 4\ \tau^2}\right] = \tfrac{1}{2}\left[\sqrt{(68.74)^2 + 4\ (40.74)^2}\right]$$
$$= 53.3 \text{ MPa}$$

Stresses at point B

We know that maximum principal (or normal) stress at point B,

$$\sigma_{B(max)} = \frac{\sigma_B}{2} + \frac{1}{2}\left[\sqrt{(\sigma_B)^2 + 4\,\tau^2}\right]$$
$$= \frac{53.46}{2} + \frac{1}{2}\left[\sqrt{(53.46)^2 + 4\,(40.74)^2}\right]$$
$$= 26.73 + 48.73 = 75.46 \text{ MPa (compressive)}$$

Minimum principal (or normal) stress at point B,

$$\sigma_{B(min)} = \frac{\sigma_B}{2} - \frac{1}{2}\left[\sqrt{(\sigma_B)^2 + 4\tau^2}\right]$$
$$= 26.73 - 48.73 = -22 \text{ MPa}$$
$$= 22 \text{ MPa (tensile)}$$

and maximum shear stress at point B,

$$\tau_{B(max)} = \tfrac{1}{2}\left[\sqrt{(\sigma_B)^2 + 4\,\tau^2}\right] = \tfrac{1}{2}\left[\sqrt{(53.46)^2 + 4\,(40.74)^2}\right]$$
$$= 48.73 \text{ MPa}$$

Theories of Failure under Static Load

- ✓ *It has already been discussed in the previous chapter that strength of machine members is based upon the mechanical properties of the materials used.*
- ✓ *These properties are usually determined from simple tension or compression tests, therefore, predicting failure in members subjected to uniaxial stress is both simple and straight-forward.*
- ✓ *But the problem of predicting the failure stresses for members subjected to bi-axial or tri-axial stresses is much more complicated.*
- ✓ *In fact, the problem is so complicated that a large number of different theories have been formulated. The principal theories of failure for a member subjected to bi-axial stress are as follows:*

  1. *Maximum principal (or normal) stress theory (also known as Rankine's theory).*

  2. *Maximum shear stress theory (also known as Guest's or Tresca's theory).*

  3. *Maximum principal (or normal) strain theory (also known as Saint Venant theory).*

  4. *Maximum strain energy theory (also known as Haigh's theory).*

  5. *Maximum distortion energy theory (also known as Hencky and Von Mises* *theory).*

Since ductile materials usually fail by yielding i.e. when permanent deformations occur in the material and brittle materials fail by fracture, therefore the limiting strength for these two classes of materials is normally measured by different mechanical properties.

Problems

The load on a bolt consists of an axial pull of 10 kN together with a transverse shear force of 5 kN. Find the diameter of bolt required according to

1. Maximum principal stress theory; 2. Maximum shear stress theory; 3. Maximum principal strain theory; 4. Maximum strain energy theory; and 5. Maximum distortion energy theory.

Take permissible tensile stress at elastic limit = 100 MPa and poisson's ratio = 0.3.

Solution. Given : $P_{t1} = 10$ kN ; $P_s = 5$ kN ; $\sigma_{t(el)} = 100$ MPa $= 100$ N/mm^2 ; $1/m = 0.3$

Let d = Diameter of the bolt in mm.

$\therefore$ Cross-sectional area of the bolt,

$$A = \frac{\pi}{4} \times d^2 = 0.7854\ d^2\ \text{mm}^2$$

We know that axial tensile stress,

$$\sigma_1 = \frac{P_{t1}}{A} = \frac{10}{0.7854\ d^2} = \frac{12.73}{d^2}\ \text{kN/mm}^2$$

and transverse shear stress,

$$\tau = \frac{P_s}{A} = \frac{5}{0.7854\ d^2} = \frac{6.365}{d^2}\ \text{kN/mm}^2$$

1. ***According to maximum principal stress theory***

We know that maximum principal stress,

$$\sigma_{t1} = \frac{\sigma_1 + \sigma_2}{2} + \frac{1}{2}\left[\sqrt{(\sigma_1 - \sigma_2)^2 + 4\tau^2}\right]$$

$$= \frac{\sigma_1}{2} + \frac{1}{2}\left[\sqrt{(\sigma_1)^2 + 4\tau^2}\right] \qquad \text{...}(\because \sigma_2 = 0)$$

$$= \frac{12.73}{2\ d^2} + \frac{1}{2}\left[\sqrt{\left(\frac{12.73}{d^2}\right)^2 + 4\left(\frac{6.365}{d^2}\right)^2}\right]$$

$$= \frac{6.365}{d^2} + \frac{1}{2} \times \frac{6.365}{d^2}\left[\sqrt{4+4}\right]$$

$$= \frac{6.365}{d^2}\left[1 + \frac{1}{2}\sqrt{4+4}\right] = \frac{15.365}{d^2}\ \text{kN/mm}^2 = \frac{15\ 365}{d^2}\ \text{N/mm}^2$$

According to maximum principal stress theory,

$$\sigma_{t1} = \sigma_{t(el)} \quad \text{or} \quad \frac{15\,365}{d^2} = 100$$

$$\therefore \quad d^2 = 15\,365/100 = 153.65 \quad \text{or} \quad d = 12.4 \text{ mm } \textbf{Ans.}$$

2. According to maximum shear stress theory

We know that maximum shear stress,

$$\tau_{max} = \tfrac{1}{2}\left[\sqrt{(\sigma_1 - \sigma_2)^2 + 4\,\tau^2}\right] = \tfrac{1}{2}\left[\sqrt{(\sigma_1)^2 + 4\,\tau^2}\right] \qquad ...(\because \sigma_2 = 0)$$

$$= \frac{1}{2}\left[\sqrt{\left(\frac{12.73}{d^2}\right)^2 + 4\left(\frac{6.365}{d^2}\right)^2}\right] = \frac{1}{2} \times \frac{6.365}{d^2}\left[\sqrt{4+4}\right]$$

$$= \frac{9}{d^2} \text{ kN/mm}^2 = \frac{9000}{d^2} \text{ N/mm}^2$$

According to maximum shear stress theory,

$$\tau_{max} = \frac{\sigma_{t(el)}}{2} \quad \text{or} \quad \frac{9000}{d^2} = \frac{100}{2} = 50$$

$$\therefore \quad d^2 = 9000 / 50 = 180 \quad \text{or} \quad d = 13.42 \text{ mm } \textbf{Ans.}$$

3. According to maximum principal strain theory

We know that maximum principal stress,

$$\sigma_{t1} = \frac{\sigma_1}{2} + \frac{1}{2}\left[\sqrt{(\sigma_1)^2 + 4\,\tau^2}\right] = \frac{15\,365}{d^2}$$

and minimum principal stress,

$$\sigma_{t2} = \frac{\sigma_1}{2} - \frac{1}{2}\left[\sqrt{(\sigma_1)^2 + 4\,\tau^2}\right]$$

$$= \frac{12.73}{2\,d^2} - \frac{1}{2}\left[\sqrt{\left(\frac{12.73}{d^2}\right)^2 + 4\left(\frac{6.365}{d^2}\right)^2}\right]$$

$$= \frac{6.365}{d^2} - \frac{1}{2} \times \frac{6.365}{d^2}\left[\sqrt{4+4}\right]$$

$$= \frac{6.365}{d^2}\left[1 - \sqrt{2}\right] = \frac{-\,2.635}{d^2} \text{ kN/mm}^2$$

$$= \frac{-\,2635}{d^2} \text{ N/mm}^2$$

We know that according to maximum principal strain theory,

$$\frac{\sigma_{t1}}{E} - \frac{\sigma_{t2}}{mE} = \frac{\sigma_{t(el)}}{E} \quad \text{or} \quad \sigma_{t1} - \frac{\sigma_{t2}}{m} = \sigma_{t(el)}$$

$$\therefore \quad \frac{15\,365}{d^2} + \frac{2635 \times 0.3}{d^2} = 100 \quad \text{or} \quad \frac{16\,156}{d^2} = 100$$

$$d^2 = 16\,156 / 100 = 161.56 \quad \text{or} \quad d = 12.7 \text{ mm } \textbf{Ans.}$$

4. *According to maximum strain energy theory*

We know that according to maximum strain energy theory,

$$(\sigma_{t1})^2 + (\sigma_{t2})^2 - \frac{2\,\sigma_{t1} \times \sigma_{t2}}{m} = [\sigma_{t(el)}]^2$$

$$\left[\frac{15\,365}{d^2}\right]^2 + \left[\frac{-\,2635}{d^2}\right]^2 - 2 \times \frac{15\,365}{d^2} \times \frac{-\,2635}{d^2} \times 0.3 = (100)^2$$

$$\frac{236 \times 10^6}{d^4} + \frac{6.94 \times 10^6}{d^4} + \frac{24.3 \times 10^6}{d^4} = 10 \times 10^3$$

$$\frac{23\,600}{d^4} + \frac{694}{d^4} + \frac{2430}{d^4} = 1 \quad \text{or} \quad \frac{26\,724}{d^4} = 1$$

$\therefore$ $d^4 = 26\,724$ or $d = 12.78$ mm **Ans.**

5. *According to maximum distortion energy theory*

According to maximum distortion energy theory,

$$(\sigma_{t1})^2 + (\sigma_{t2})^2 - 2\sigma_{t1} \times \sigma_{t2} = [\sigma_{t(el)}]^2$$

$$\left[\frac{15\,365}{d^2}\right]^2 + \left[\frac{-\,2635}{d^2}\right]^2 - 2 \times \frac{15\,365}{d^2} \times \frac{-\,2635}{d^2} = (100)^2$$

$$\frac{236 \times 10^6}{d^4} + \frac{6.94 \times 10^6}{d^4} + \frac{80.97 \times 10^6}{d^4} = 10 \times 10^3$$

$$\frac{23\,600}{d^4} + \frac{694}{d^4} + \frac{8097}{d^4} = 1 \quad \text{or} \quad \frac{32\,391}{d^4} = 1$$

$\therefore$ $d^4 = 32\,391$ or $d = 13.4$ mm **Ans.**

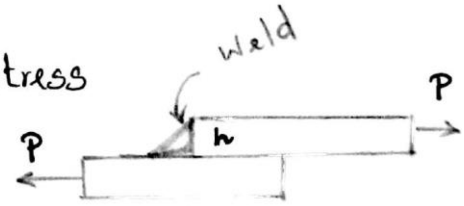
Welding . Unit-I

Strength of Transverse Fillet Welded joints :

For Single Fillet Weld :

$$P = \text{Throat Area} \times \text{Allowable tensile stress}$$

$$P = 0.707\,h \times l \times \sigma_t$$

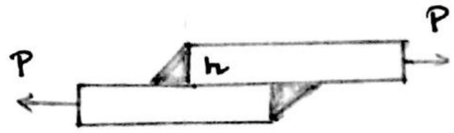


Single

For Double Fillet Weld :

$$P = 2 \times 0.707\,h \times l \times \sigma_t$$

$$P = 1.414\,h \times l \times \sigma_t$$



Double.

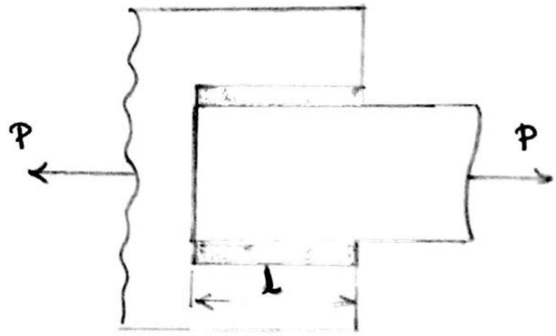
Strength of parallel fillet Welded joints :

For Single Parallel Weld :

$$P = 0.707\,h \times l \times \tau$$

For Double Parallel Weld :

$$P = 1.414\,h \times l \times \tau$$



If there is Combination of single transverse and double parallel fillet Weld :

$$P = 0.707\,h \times l_1 \times \sigma_t + 1.414\,h \times l_2 \times \tau$$

※. In Welding We have to add 12.5 mm for starting and stopping of Weld run.

Stress Concentration factor for Welded joints.

Type of joint	Stress Concentration Factor
Transverse fillet Welds	1.5
Parallel fillet Welds	2.7

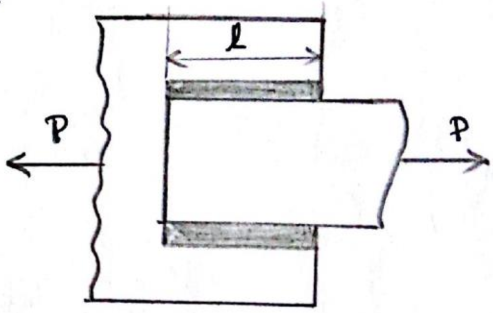
A plate 100 mm wide and 12.5 mm thick is to be welded to another plate by means of parallel fillet welds. The plates are subjected to a load of 50 KN. Find the length of the weld so that the maximum stress does not exceed 56 MPa. Consider the joint first under static loading and then under fatigue loading.

Given: Width (b) = 100 mm; Thickness t (or) h = 12.5 mm;

$P = 50\ KN = 50 \times 10^3\ N$; $\tau = 56\ MPa$ (or) $= 56\ N/mm^2$;

Length of weld for static loading:

W.K.T, the maximum load which the plates can carry for double parallel fillet welds.



$$P = 1.414\, h \times l \times \tau$$

$$50 \times 10^3 = 1.414 \times 12.5 \times l \times 56$$

$$l = \frac{50 \times 10^3}{990} = 50.5\ mm$$

Add 12.5 mm for starting and stopping of weld run

$$l = 50.5 + 12.5$$

$$l = 63\ mm$$

Length of weld for fatigue loading:

$$\text{Permissible shear stress} = \frac{\text{Allowable shear stress}}{\text{Stress concentration factor}}$$

$$\tau = \frac{56}{2.7} = 20.74\ N/mm^2$$

W.K.T

$$P = 1.414 \times h \times l \times \tau$$

$$50 \times 10^3 = 1.414 \times 12.5 \times l \times 20.74$$

$$l = 136.2\ mm$$

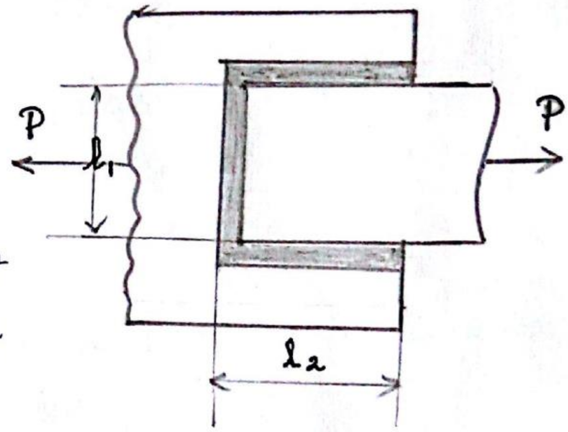
Add 12.5 mm

$$l = 136.2 + 12.5$$

$$l = 148.7\ mm.$$

A plate 75 mm Wide and 12.5 mm thick is joined With another Plate by a single transverse Weld and a double parallel fillet Weld as shown in fig. The maximum tensile and shear stresses are 70 MPa and 56 MPa respectively.

Find the Length of each Parallel fillet, if the joint is Subjected to both static and fatigue Loading.



Given :

Width (b) $= 75$ mm

Thickness (h) $= 12.5$ mm

$\sigma_t = 70$ MPa or N/mm^2

$\tau = 56$ MPa or N/mm^2

Where

l_1 = Length of the transverse Weld.

l_2 = Length of the parallel fillet Weld.

For l_1 We have to Subtract 12.5 mm from the Width of the plate

$$l_1 = \text{Width} - 12.5 \text{ mm}$$

$$l_1 = 75 - 12.5$$

$$l_1 = 62.5 \text{ mm.}$$

Length of each parallel fillet for static Loading.

W.K.T

$$P = P_1 + P_2$$

Where

P = Maximum Load on the plate

P_1 = Load Carried by transverse Weld

P_2 = Load Carried by Parallel Weld

$$\sigma = \frac{\text{Load}}{\text{Area}} = \frac{P}{A}$$

$$\text{Area} = \text{Width} \times \text{Thickness}$$

$$A = 75 \times 12.5 = 937.5 \text{ mm}^2$$

Where $\sigma = 70\ N/mm^2$

$P = 70 \times 937.5$

$P = 65625\ N$

$P_1 = 0.707\ h \times l_1 \times \sigma_t$ → Single transverse Weld

$= 0.707 \times 12.5 \times 62.5 \times 70$

$P_1 = 38664\ N$

$P_2 = 1.414 \times h \times l_2 \times \tau$ → Double Parallel Weld.

$= 1.414 \times 12.5 \times l_2 \times 56$

$P_2 = 990\ l_2\ N$

∴ Load Carried by the joint

$P = P_1 + P_2$

$65625 = 38664 + 900\ l_2$

$l_2 = 27.2\ mm.$

Add 12.5 mm for Weld run.

$l_2 = 27.2 + 12.5$

$= 39.7\ mm$

$l_2 \simeq 40\ mm.$

Length of each parallel fillet for fatigue Loading:

Stress Concentration factor for Transverse & parallel are 1.5 & 2.7.

So, $\sigma_t = 70/1.5 = 46.7\ N/mm^2$

$\tau = 56/2.7 = 20.74\ N/mm^2$

$P_1 = 0.707 \times 12.5 \times 62.5 \times 46.7 = 25795\ N$

$P_2 = 1.414 \times 12.5 \times l_2 \times 20.74 = 366\ l_2\ N$

∴ $65625 = 25795 + 366\ l_2$ → $P = P_1 + P_2$

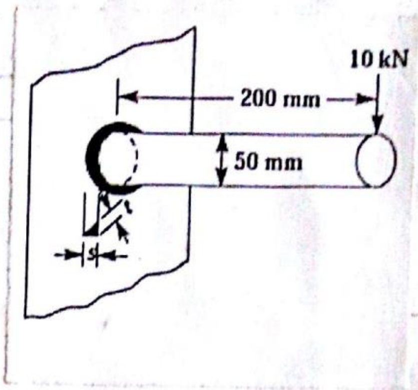
$l_2 = 108.8$

Add 12.5 mm for Weld run,

$l_2 = 108.8 + 12.5$

$l_2 = 121.3\ mm$

A 50 mm diameter Solid shaft is Welded to a flat plate as shown in figure. If the Size of the Weld is 15 mm, find the maximum normal stress and shear stress in the Weld.



Given :

$D = 50$ mm

$h = 15$ mm

$L = 200$ mm

$P = 10$ kN $= 10 \times 10^3$ N.

Maximum Normal stress :

$$\sigma_{max} = \frac{1}{2}\sigma_b + \frac{1}{2}\sqrt{\sigma_b^2 + 4\tau^2}$$ Pag: No: 7.2

Maximum shear stress :

$$\tau_{max} = \frac{1}{2}\sqrt{\sigma_b^2 + 4\tau^2}$$ Pag: No: 7.2

$$\sigma_b = \frac{5.66\, M_b}{h D^2 \pi}$$ [M_b is Bending Moment] Pag: No: 11.3

$M_b = P \times L$

$$\sigma_b = \frac{5.66 \times 10 \times 10^3 \times 200}{15 \times 50^2 \times \pi}$$

$\sigma_b = 96$ N/mm^2 or MPa.

$$\tau = \frac{2.83\, M_t}{h \cdot D^2 \cdot \pi}$$ [M_t is twisting Moment] Pag: No: 11.3

$M_t = P \times R$ $\quad$ M_t = Load x Radius

$$\tau = \frac{2.83 \times 10 \times 10^3 \times \cancel{200}\ 25}{15 \times 50^2 \times \pi}$$

$\tau = 6$ N/mm^2 or 6 MPa

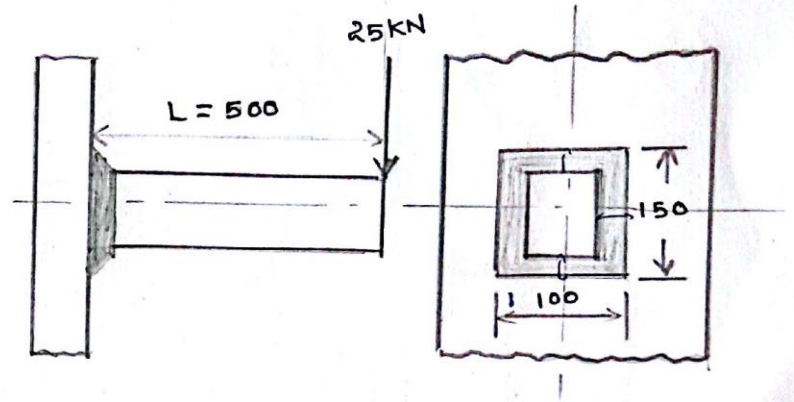
Max. Normal stress

$$\sigma_{max} = \frac{1}{2} \times 96 + \frac{1}{2}\sqrt{96^2 + 4 \times 6^2} = 96.4 \text{ MPa}$$

Max. shear stress

$$\tau_{max} = \frac{1}{2}\sqrt{96^2 + 4 \times 6^2} = 48.4 \text{ MPa}.$$

A rectangular Cross-section bar is Welded to a Support by means of fillet Welds as shown in figure. Determine the Size of the Welds, if the Permissible Shear stress in the Weld to 75 MPa



Given: $P = 25\ kN = 25 \times 10^3\ N$

$\tau_{max} = 75\ MPa$

$b = 100\ mm$

$d = 150\ mm$

$L = 500\ mm$

W.K.T

$$\tau_{max} = \frac{1}{2}\sqrt{\sigma_b^2 + 4\tau^2}$$

Pag: No: 7.2

shear stress $\tau = \frac{P}{A}$

$t = 0.707S$

Area $= t(2b + 2d)$

$= 0.707S\,(2 \times 100 + \cancel{2 \times 150}\ 2 \times 150)$

$[S = h]$

$A = 353.5\,S\ mm^2$

$$\tau = \frac{25 \times 10^3}{353.5S} = \frac{70.72}{S}$$

Bending stress $\sigma_b = \frac{M_b}{Z}$

$M = P \times L$

Z is section Modulus

$$z = t\left[bd + \frac{d^2}{3}\right]$$ → Pag: No: 11.6

$$= 0.707S\left[150 \times 100 + \frac{150^2}{3}\right]$$

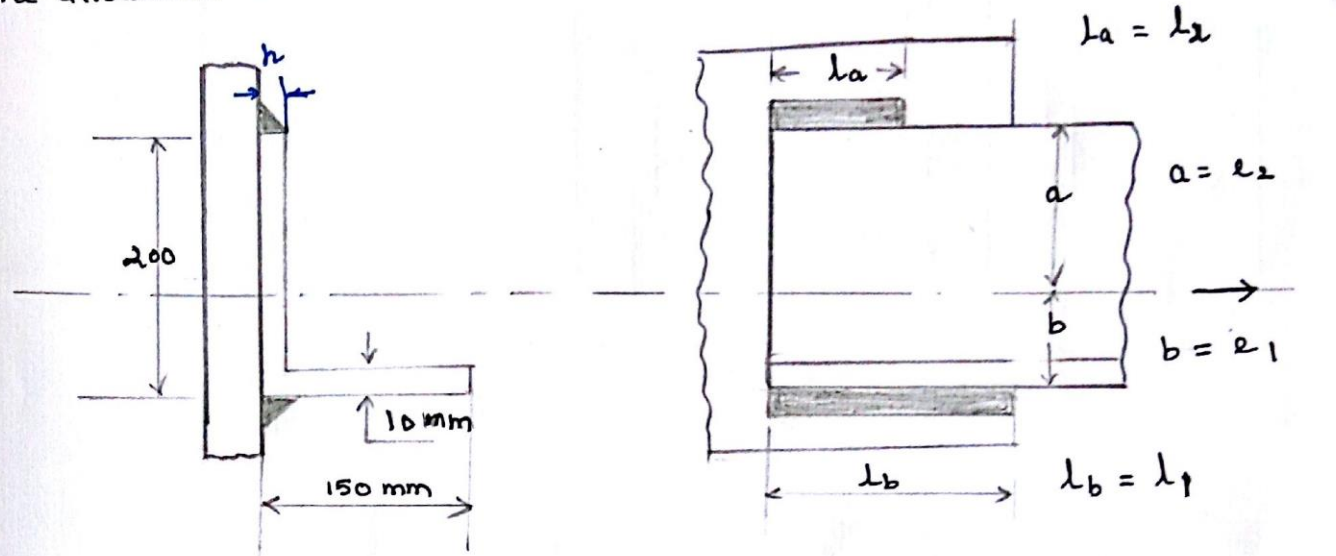
$= 15907.5\,S\ mm^3$

$$\sigma_b = \frac{25 \times 10^3 \times 500}{15907.5S} = \frac{785.8}{S}$$

$$75 = \frac{1}{2}\sqrt{\left(\frac{785.8}{S}\right)^2 + 4\left(\frac{70.72}{S}\right)^2} = \frac{399.2}{S}$$

h (or) $S = 5.32\ mm.$

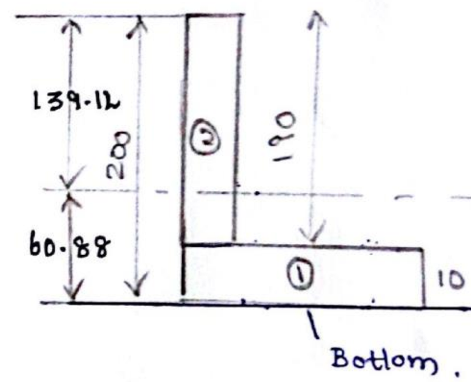
A 200 x 150 x 10 mm angle is to be Welded to a Steel Plate by fillet Welds as shown in figure. If the angle is subjected to a static Load of 200 KN, find the Length of Weld at the top and bottom. The allowable shear stress for static Loading may be taken as 75 MPa.



Given ; $P = 200\ KN = 200 \times 10^3$

$\tau = 75\ MPa$

$$l_1 = \frac{1.414\ P\ e_2}{\sigma\ h\ b} \qquad l_2 = \frac{1.414\ P\ e_1}{\sigma\ h\ b}$$



Centroid

$$\bar{y}_{bottom} = \frac{A_1 y_1 + A_2 y_2}{A_1 + A_2}$$

$A_1 = 10 \times 150 = 1500$

$y_1 = \frac{10}{2} = 5$

$A_2 = 190 \times 10 = 1900$

$y_2 = \frac{190}{2} + 10 = 105$

$$\bar{y}_{botlom} = \frac{1500 \times 5 + 1900 \times 105}{1500 + 1900} = 60.88\ mm$$

$\bar{y}_{top} = 200 - 60.88 = 139.12$

$$l_1 = \frac{1.414 \times 200 \times 10^3 \times 139.12}{75 \times 10 \times 200} \qquad l_2 = \frac{1.414 \times 200 \times 10^3 \times 60.88}{75 \times 10 \times 200}$$

$l_1 = 262.28\ mm$ (or) l_b $\qquad$ $l_2 = 114.7\ mm$ (or) l_a